

Arpandeeep Singh

602-880-3175 | [Email](#) | [Linkedin](#) | [Portfolio](#)

EDUCATION

Arizona State University

Bachelor of Science in Computer Science | GPA: 4.0

Tempe, AZ

Expected: May 2026

EXPERIENCE

Software Engineer Intern | Atya Labs | Tempe, AZ

May 2024 – Present

- Designed and implemented a backend document retrieval service supporting 1000+ documents per workspace, using vector indexing, request caching, and access-controlled APIs to deliver low-latency semantic search.
- Built asynchronous analytics pipelines integrating Google Search Console and Analytics APIs, using batched workers and rate-limit-aware retries to transform raw metrics into structured reports.
- Developed a full-stack analytics dashboard with React, REST APIs, and Firebase-backed persistence, supporting multi-role access and reducing manual onboarding and reporting effort for internal teams.

Software Developer Intern | VB Cosmetics Inc. | Chandler, AZ

May 2025 – Aug 2025

- Designed and implemented Java-based backend services to process 15K+ QA documents using validated workflows and batch processing, reducing manual effort by 90% and shortening end-to-end QA cycles by 85%.
- Built secure backend middleware in Python using OAuth-based authentication to integrate Amazon SP-API with SYSPRO ERP, eliminating recurring manual workflows and reducing operational costs by \$12K annually.
- Developed scalable ETL pipelines to migrate 100K+ Zendesk records into Salesforce with schema validation and transformation logic, achieving 100% mapped accuracy and enabling near real-time reporting across 6 departments.
- Implemented data aggregation and integration layers supporting Power BI dashboards sourced from SYSPRO ERP, automating KPI computation and reducing financial reconciliation time by 30%.

PROJECTS

Distributed Task Queue and Job Orchestration Platform | Java, Spring Boot, Kafka, PostgreSQL, Docker

- Designed and implemented an asynchronous task queue using Kafka to decouple background job execution from user-facing request paths, improving system responsiveness.
- Implemented persistent job state management in PostgreSQL with atomic transitions for pending, completed, and failed states, enabling retries and basic fault recovery.
- Built scalable worker services to batch-consume tasks, execute jobs safely, and persist execution logs for monitoring, debugging, and failure analysis.

Real-Time Consistency and Collaboration Engine | Java, WebSockets, Redis Pub/Sub, PostgreSQL

- Developed a real-time collaboration backend using WebSockets to synchronize shared application state across multiple concurrent users with low-latency updates.
- Integrated Redis Pub/Sub to propagate state changes across server instances, ensuring consistent data delivery in a horizontally scaled environment.
- Persisted shared state in PostgreSQL and implemented basic conflict detection logic to minimize overwrites during simultaneous user updates.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript, C/C++, SQL (PostgreSQL, MySQL)

Backend & Web Frameworks: Spring Boot, FastAPI, Node.js, React, Flask

Data & Systems: Kafka, Redis, PostgreSQL, MongoDB, ETL Pipelines, WebSockets

AI Tools & Platforms: OpenAI API, Gemini API, Pinecone Vector DB, RAG Pipelines (LangChain)

Cloud & Infrastructure: AWS, Google Cloud Platform (GCP), Docker, CI/CD Pipelines, Supabase, Firebase

APIs & Integrations: RESTful APIs, OAuth 2.0, Web Push (VAPID)

Developer Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Jupyter, Power BI

LEADERSHIP

Vice President, ACEL-ASU: Directed 20+ high-impact professional and cultural events engaging 500+ attendees, advancing AAPI leadership, mentorship, and career development.

Treasurer, Startup Village: Secured \$5,000+ in ASU funding to organize large-scale entrepreneurial events, strengthening campus innovation and startup collaboration.

Undergraduate Teaching Assistant: Mentored and guided 1,000+ computer science students through instruction, labs, and assessments, improving engagement and academic outcomes.